

GUIDED LECTURE NOTES
for
PROOF, SET THEORY, AND LOGIC
MATH 330

Part III: Real Numbers and Cardinality

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These guided notes accompany the Fall 2026 revised course edition of *Elementary Set Theory: Proof, Set Theory, and Logic*. They are based on Richard P. Millspaugh's original textbook materials and lecture materials prepared and revised by Bryce A. Christopherson.

The blank spaces in the student copy are completed during class. Each blank occupies exactly the same space as the corresponding typed completion in the instructor copy, so the two packets have identical pagination. Definitions, theorem statements, and the surrounding explanations are provided so that class time can focus on examples, proof decisions, and the mathematical steps that connect them.

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CHAPTER 6

THE REAL NUMBERS

INTRODUCTION

We now turn our attention to developing a mathematical description of the real numbers. All of the results of calculus can be derived from these basic properties of the real number system, which is usually done in a first course in real analysis. Our basic assumptions about the real numbers are called axioms, and they are divided into several groups. Many of these axioms will be familiar to you from previous courses in algebra.

6.1 FIELD AXIOMS

Taken as a group, the field axioms tell us that the real numbers together with the operations of addition and multiplication form what is known as a *field*. Very informally, you might think of a field as something that satisfies the usual rules you encountered in high school algebra. Fields and related objects are studied in more depth in a course in abstract algebra.

Field Axioms.

The set of real numbers $\mathbb{R}$ is a collection of objects together with the two binary operations addition and multiplication that satisfy the following properties (known as the field axioms):

- (i) Commutative Laws: For every $a, b \in \mathbb{R}$, $a + b = b + a$ and $ab = ba$.
- (ii) Associative Laws: For every $a, b, c \in \mathbb{R}$, $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.
- (iii) Distributive Law: For every $a, b, c \in \mathbb{R}$, $a(b + c) = ab + ac$.
- (iv) Identities: There are distinct elements 0 and 1 in $\mathbb{R}$ such that $a \cdot 1 = a$ and $a + 0 = a$ for every $a \in \mathbb{R}$.
- (v) Additive Inverses: For every $a \in \mathbb{R}$, there is an element $-a \in \mathbb{R}$ such that $a + (-a) = 0$.
- (vi) Multiplicative Inverses: For every $a \in \mathbb{R}$ with $a \neq 0$, there is an element $a^{-1} \in \mathbb{R}$ such that $aa^{-1} = 1$.

Any set of objects satisfying all of these properties is a *field*. The rational numbers $\mathbb{Q}$ and complex numbers $\mathbb{C}$ are also fields. The additional order and completeness axioms below distinguish $\mathbb{R}$ from these examples.

A number of the familiar algebraic properties of the real numbers are immediate consequences of the field axioms. A few of them are collected in the following theorem, though there are certainly many others.

THEOREM 6.1.

For any real numbers a, b, c , the following are true:

- (i) *If $a + b = a + c$, then $b = c$.*
- (ii) *$-(-a) = a$.*
- (iii) *If $a \neq 0$ and $ab = ac$, then $b = c$.*
- (iv) *If $a \neq 0$, then $(a^{-1})^{-1} = a$.*
- (v) *$a \cdot 0 = 0$.*
- (vi) *$-a = (-1)a$.*
- (vii) *If $ab = 0$, then $a = 0$ or $b = 0$.*

Proof.

6.2 ORDER AXIOMS

As noted previously, there are a number of common fields. One of the differences between $\mathbb{R}$ and $\mathbb{C}$ is that we think of the real numbers as lying in order along a line. There is no natural way to organize the complex numbers in this fashion. The next group of axioms make precise what we mean when we say that the real numbers form an *ordered field*.

Order Axioms. There is an order $<$ defined on the real numbers $\mathbb{R}$ satisfying:

(i) Transitivity: If $a < b$ and $b < c$, then $a < c$.

(ii) Trichotomy: For every two real numbers a and b , exactly one of the following holds:

$$a < b \quad \text{or} \quad a = b \quad \text{or} \quad b < a$$

(iii) If $a < b$, then $a + c < b + c$ for every $c \in \mathbb{R}$.

(iv) If $a < b$ and $c > 0$, then $ac < bc$.

We derive several consequences of the fact that $\mathbb{R}$ is an ordered field.

THEOREM 6.2.

The following statements are true in $\mathbb{R}$:

- (i) $a > 0$ iff $-a < 0$.
- (ii) If $a < b$, then $-a > -b$.
- (iii) If $a \neq 0$, then $a^2 > 0$.
- (iv) $1 > 0$.

Proof.

THEOREM 6.3.

Let $a, b \in \mathbb{R}$.

- (i) If $a > 0$ and $b > 0$, then $ab > 0$.
- (ii) If $a < 0$ and $b < 0$, then $ab > 0$.
- (iii) If $a > 0$ and $b < 0$, then $ab < 0$.

THEOREM 6.4.

Let $a \in \mathbb{R}$.

- (i) If $a > 0$, then $a^{-1} > 0$.
- (ii) If $a < 0$, then $a^{-1} < 0$.

Proof.

THEOREM 6.5.

Let $a \geq 0$ and $b \geq 0$. Then $a < b$ iff $a^2 < b^2$.

Proof.

6.3 COMPLETENESS OF THE REAL NUMBERS

Our axioms so far ensure that $\mathbb{R}$ is an ordered field. The field $\mathbb{Q}$ of rational numbers is also an ordered field, so we need something further to distinguish between $\mathbb{R}$ and $\mathbb{Q}$. Our last axiom is based on the observation that the field $\mathbb{Q}$ has *holes*, whereas $\mathbb{R}$ does not. Let's first consider an example to clarify what we mean by this.

EXAMPLE 6.6.

6.3.1 UPPER AND LOWER BOUNDS

DEFINITION 6.7. Let A be a nonempty set of real numbers. We say that a number u is an *upper bound* for A if $x \leq u$ for every $x \in A$. We say that a number m is a *lower bound* for A if $x \geq m$ for every $x \in A$. We say that A is *bounded above* if A has an upper bound, that A is *bounded below* if A has a lower bound, and that A is *bounded* if A is bounded above and below.

EXAMPLE 6.8.

The previous example illustrates several things. It is possible for a nonempty set to have both upper and lower bounds, just one bound, or no bounds at all. Upper and lower bounds may or may not be elements of the set. Finally, upper and lower bounds are not unique. Transitivity implies that if M is an upper bound for a set A , then every number larger than M is also an upper bound for A . Similarly, if m is a lower bound for A then every number smaller than m is a lower bound for A .

Consider the set $A = \{a \in \mathbb{R} \mid a^2 \leq 2\}$ from the previous example again. We claimed that 2 is an upper bound, and that is certainly true, but note that $3/2$ is also an upper bound. In some sense this smaller upper bound is a “better” bound for A because it puts a tighter restriction on the size of the elements of A . This is approximately like saying that Grand Forks is a better way to describe the location of UND than North Dakota. In this sense the best possible upper bound would be the smallest one, if there is one. In this case, A has a smallest upper bound in $\mathbb{R}$, namely $\sqrt{2}$. The analogous rational set $\{q \in \mathbb{Q} \mid q^2 \leq 2\}$ has no least upper bound in $\mathbb{Q}$. This is the basic difference between $\mathbb{Q}$ and $\mathbb{R}$ that we wish to capture in an axiom. First, we need a couple more definitions.

DEFINITION 6.9. Let A be a nonempty subset of $\mathbb{R}$. We say that a number u is a *least upper bound* (LUB) for A if both of the following are true:

- (i) For every $a \in A$, $a \leq u$.
- (ii) For every upper bound b of A , $u \leq b$.

We say that a number m is a *greatest lower bound* (GLB) for A if both of the following are true:

- (i) For every $a \in A$, $a \geq m$.
- (ii) If b is a lower bound for A , then $m \geq b$.

Note that a LUB is sometimes called a *supremum* and a GLB is sometimes called an *infimum*.

It is not hard to prove that, unlike upper bounds, a set can have only one least upper bound. This fact is made explicit in the following theorem, whose proof is left as a homework exercise.

THEOREM 6.10.

Let A be a nonempty subset of $\mathbb{R}$. If u and v are least upper bounds for A , then $u = v$.

The following theorem says that the least upper bound of a set must in some sense be “close to” the set.

THEOREM 6.11.

Let A be a nonempty subset of $\mathbb{R}$ and let u be the least upper bound for A . If $x < u$, then there is an element $a \in A$ such that $x < a$.

Proof.

We are now ready to state our final axiom, which says that $\mathbb{R}$ is *complete*.

The Completeness Axiom. Let A be a nonempty subset of $\mathbb{R}$. If A has an upper bound, then A has a least upper bound.

There are a number of consequences of the Completeness Axiom, most of which are beyond the scope of this text. We will look at only a couple of them. We begin with the fairly intuitive-seeming fact that for any real number x , there is a natural number larger than x .

THEOREM 6.12.

(Archimedean Property) If $x \in \mathbb{R}$, then there is an $n \in \mathbb{N}$ such that $n > x$.

Proof.

We previously commented, without proof, that the Completeness Axiom would allow us to distinguish $\mathbb{R}$ from $\mathbb{Q}$. We will now make this explicit by proving that there is at least one real number that is not rational.¹ By Theorem 2.18, there is no rational number whose square is 2. We now show that the Completeness Axiom implies that there must be a real number x with $x^2 = 2$.

THEOREM 6.13.

There is a number $x \in \mathbb{R}$ such that $x^2 = 2$.

Proof.

¹In fact there are more irrational numbers than there are rational numbers, as we shall see in the final chapter.

CHAPTER 7

INTRODUCTION TO CARDINALITY

INTRODUCTION

What do we mean when we say that there are *four* suits in a standard deck of cards? More generally, what does it mean to count any collection of objects? Since you may no longer actually think about counting, it may help to watch a child who is just learning to count. Given four objects to count, the child is likely to point to each object in turn and count “one, two, three, four.” In other words, the child is explicitly constructing a bijective function between the objects she is counting and the elements of the set $\{1, 2, 3, 4\}$. Our goal in this chapter is to extend this idea to infinite sets.

7.1 THE CARDINALITY OF A SET

For sets A and B , write $A \equiv B$ if there is a bijective function $f : A \rightarrow B$. In this case we say that A and B are *equinumerous* or that they have the same *cardinality*. Write $A \leq B$ if there is an injective function $f : A \rightarrow B$; equivalently, $B \geq A$. We write $A < B$ to indicate that $A \leq B$ and $A \not\equiv B$. It is also common to write $|A| = |B|$ rather than $A \equiv B$.

THEOREM 7.1.

For any set $\mathcal{C}$ whose elements are sets, equinumerosity restricts to an equivalence relation on $\mathcal{C}$.

Proof.

Note that Theorem 7.1 allows us to say that two sets A and B have the same cardinality if we are able to find a bijection from A to B or a bijection from B to A .

THEOREM 7.2.

On any set $\mathcal{C}$ whose elements are sets, the relation $\leq$ is reflexive and transitive.

Proof.

While we would not expect $\leq$ to be symmetric, we can ask for a cardinal analogue of antisymmetry. Mutual injections do not force two sets to be literally equal, but do they force the sets to be equinumerous? Georg Cantor (1845–1918) was interested in this question, and Felix Bernstein (1878–1956) proved that the answer is yes. The result is usually known as the Cantor–Bernstein Theorem.

THEOREM 7.3 (Cantor–Bernstein).

If $A \leq B$ and $B \leq A$, then $A \equiv B$.

We prove this theorem in Chapter A. The following example uses the same kind of chain-shifting idea to construct a bijection between the open interval $(-1, 1)$ and the closed interval $[-1, 1]$.

EXAMPLE 7.4.

7.2 FINITE SETS

QUESTION.

What does it mean to say that a set A is finite?

At first glance, this may seem like something you've known for a long time. Don't we just mean that we can count the elements of A ? If so, is the number of cells in your body finite? Can you count them? Maybe the answer to our question isn't quite so obvious after all.

For each $n \in \mathbb{N} \cup \{0\}$, define

$$I_0 = \emptyset \quad \text{and} \quad I_n = \{1, 2, \dots, n\} \quad \text{when } n \in \mathbb{N}.$$

Thus $I_4 = \{1, 2, 3, 4\}$. These sets let us formalize the idea of counting while handling the empty set without a separate convention.

DEFINITION 7.5.

Let A be a set and let $n \in \mathbb{N} \cup \{0\}$. We say that A has n elements if $A \equiv I_n$. The set A is *finite* if $A \equiv I_n$ for some $n \in \mathbb{N} \cup \{0\}$, and it is *infinite* if it is not finite.

THEOREM 7.6. *The set $\mathbb{N}$ is infinite.*

Proof.

Composing a bijection $A \rightarrow I_n$ with the inclusion $I_n \hookrightarrow \mathbb{N}$ gives the following.

THEOREM 7.7.

If A is finite, then $A \leq \mathbb{N}$.

Now that we have a definition of finite, let's reconsider the set C of cells in your body. Is C finite? If so, for which n is $C \equiv I_n$? We still can't really count them, so perhaps we've just obscured the question rather than answering it. Scientists estimate that there are about 10,000,000,000,000 cells in the average adult human body. That's not an actual count of

the number of cells in any individual human body, though. These kinds of estimates are based on the sizes of various kinds of cells and the approximate proportion of each kind of cell in the body. In fact, we could determine the maximum number of cells that might be in a person's body by figuring out how many of the smallest kinds of cells would be required to build a body of a particular volume, or weight, etc. It seems reasonable to think that we could say a set was finite if we were sure it had at most n elements for some natural number n . That is the intent of the next result.

THEOREM 7.8.

If $A \leq I_n$ for some $n \in \mathbb{N} \cup \{0\}$, then A is finite.

Suppose there is an injection $f : A \rightarrow I_n$. Our definition of finite asks for a bijection with some I_k , while f need not be surjective. The next lemma removes one unused point from the codomain at a time.

LEMMA 7.9.

Let $n \in \mathbb{N}$. If $f : A \rightarrow I_n$ is injective but not surjective, then there is an injection $g : A \rightarrow I_{n-1}$.

Proof.

We will now prove the theorem.

Proof of Theorem 7.8.

Consequently, if $\mathbb{N} \leq A$, then A is infinite.

We conclude this section with a question, the answer to which may seem obvious to you.

QUESTION 7.10.

If $m, n \in \mathbb{N} \cup \{0\}$ and $m \neq n$, can you prove that $I_m \not\cong I_n$?

7.3 DENUMERABLE SETS

Georg Cantor was able to define a complete system of infinite numbers and of arithmetic on those numbers. We will not discuss his system here, but we will look at one particular kind of infinite number that is important in many areas of mathematics.

THEOREM 7.11.

If A is an infinite set and B is a finite subset of A , then the set $A \setminus B$ is infinite.

Proof.

DEFINITION 7.12.

Let A be a set. We say that A is:

- *denumerable* if $A \equiv \mathbb{N}$.
- *countable* if A is either finite or denumerable.
- *uncountable* if A is infinite and not denumerable.

EXAMPLE 7.13.

The preceding example points out an important difference between finite and infinite sets. The set A is a proper subset of $\mathbb{N}$, but has the same cardinality as $\mathbb{N}$. A set equinumerous with one of its proper subsets is called *Dedekind-infinite*. To see why a finite set cannot have this property, suppose $A \equiv I_n$ and $B \subsetneq A$. The image of B under a bijection $A \rightarrow I_n$ is a proper subset S of I_n ; in particular, $n \geq 1$. The inclusion $S \rightarrow I_n$ is not surjective, so Lemma 7.9 gives an injection $S \rightarrow I_{n-1}$. If $B \equiv A$, this would produce an injection $I_n \rightarrow I_{n-1}$, contradicting the induction in the completion of Question 7.10. Therefore every Dedekind-infinite set is infinite. The converse holds in standard ZFC set theory, but its proof uses a form of the Axiom of Choice; without Choice, the two notions need not coincide.

We would like to determine which of our results about finite sets are also true for denumerable sets. If A and B are denumerable, must $A \cup B$ also be denumerable? Are subsets of denumerable sets denumerable? Is there an analog of Theorem 7.8? Are there other ways to tell that a set is denumerable?

THEOREM 7.14.

If A is a denumerable set and $B \equiv A$, then B is denumerable.

Proof. By definition we have $A \equiv \mathbb{N}$. Since $\equiv$ is an equivalence relation, it follows that $B \equiv \mathbb{N}$ as desired.

THEOREM 7.15.

If $A \subseteq \mathbb{N}$, then A is countable.

Proof.

COROLLARY 7.16. *A set A is countable if and only if $A \leq \mathbb{N}$.*

Proof.

COROLLARY 7.17. *A subset of a countable set is countable.*

Proof.

Thus a set A is countable exactly when there is an injection from A to $\mathbb{N}$. If A is nonempty, this is also equivalent to finding a surjection from $\mathbb{N}$ onto A . In one direction, invert an injection on its range and send unused natural numbers to a fixed element of A ; in the other, assign each $a \in A$ the least natural number mapping to it. The nonempty hypothesis is necessary: there is no function from $\mathbb{N}$ onto $\emptyset$.

COROLLARY 7.18.

A set A is countable if either of the following is true:

- (i) There is an injection $f : A \rightarrow \mathbb{N}$.*
- (ii) $A \neq \emptyset$ and there is a surjection $f : \mathbb{N} \rightarrow A$.*

THEOREM 7.19.

The union of two denumerable sets is denumerable.

Proof.

Using Mathematical Induction on the number of sets, we obtain:

COROLLARY 7.20.

The union of any positive finite number of denumerable sets is denumerable.

COROLLARY 7.21. *The union of two countable sets is countable.*

Proof.

THEOREM 7.22.

The set $\mathbb{N} \times \mathbb{N}$ is denumerable.

Proof.

7.3.1 THE RATIONAL NUMBERS

At first glance it may seem that there are more rational numbers than there are natural numbers. After all, there are infinitely many rational numbers between any two natural numbers. One of Cantor's accomplishments was to show that the set of rational numbers is actually denumerable. Our intuition developed from years of working with finite sets just doesn't serve us very well when working with infinite sets.

LEMMA 7.23.

The set $\mathbb{Q}^+$ of positive rational numbers is countable.

Proof.

Since the function taking every positive rational to its additive inverse is bijective, the following corollary is immediate from the lemma.

COROLLARY 7.24.

The set $\mathbb{Q}^-$ of negative rational numbers is countable.

We know that $\mathbb{Q}^+$, $\mathbb{Q}^-$, and $\{0\}$ are all countable sets. Applying Corollary 7.21 twice, we conclude:

THEOREM 7.25.

The set $\mathbb{Q}$ of rational numbers is denumerable.

Proof.

7.3.2 THE REAL NUMBERS

By this point it may seem possible that all sets are countable, making denumerability a useless distinction. We will show that this is not true by proving that the real numbers are uncountable. A real number can have two decimal expansions; for example, $1.000 \dots = 0.999 \dots$. To make decimal digits unambiguous, we choose for each real number its *canonical* expansion: the one that is not eventually all 9s.

THEOREM 7.26.

The set $\mathbb{R}$ of real numbers is uncountable.

Proof.

APPENDIX A

THE CANTOR–BERNSTEIN THEOREM

The relation $X \preceq Y$ means that X injects into Y . It is reflexive and transitive, but it is not literally antisymmetric on sets: distinct sets can have the same cardinality. The Cantor–Bernstein Theorem supplies the appropriate cardinal version of antisymmetry:

$$X \preceq Y \text{ and } Y \preceq X \implies X \equiv Y.$$

The proof below is constructive once the two injections are given.

A.1 THE FORWARD-CHAIN CONSTRUCTION

Assume that $f : X \rightarrow Y$ and $g : Y \rightarrow X$ are injective. Some points of X have no predecessor under g ; begin a forward chain at each such point. Define

$$A_0 = X \setminus g[Y], \quad A_{n+1} = g[f[A_n]] \quad (n = 0, 1, 2, \dots),$$

and let

$$A = \bigcup_{n=0}^{\infty} A_n = \{x \in X \mid x \in A_n \text{ for some } n \in \mathbb{N} \cup \{0\}\}.$$

On the chains in A we use f in the forward direction. Off those chains, every point has a unique predecessor under g , so we use that predecessor.

Proof of Theorem 7.3.

A.2 WHY THE CONSTRUCTION WORKS

The set A_0 records where the backward search through g must stop. Each A_{n+1} advances those starting points one step through f and then g . Using f on every such forward chain prevents a missing predecessor from causing a gap. Everywhere else, the g -chains extend backward indefinitely, so reversing g is safe. The argument also covers empty sets without a separate case.